```
[3]
✓ 0s
```

```python
import numpy as np
import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression

# Dataset
X = np.array([[1], [2], [3], [4], [5]])
y = np.array([2, 4, 6, 8, 10])

# Create model
model = LinearRegression()

# Train model
model.fit(X, y)

# Prediction
y_pred = model.predict(X)

# New prediction
new_prediction = model.predict([[6]])

print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)

print("Prediction for X = 6:")
print(new_prediction[0])

# Graph
plt.figure(figsize=(8, 5))

plt.scatter(
    X,
    y,
    s=80,
```

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Go to Settings to activate Windows.

```
[3]
✓ 0s
```

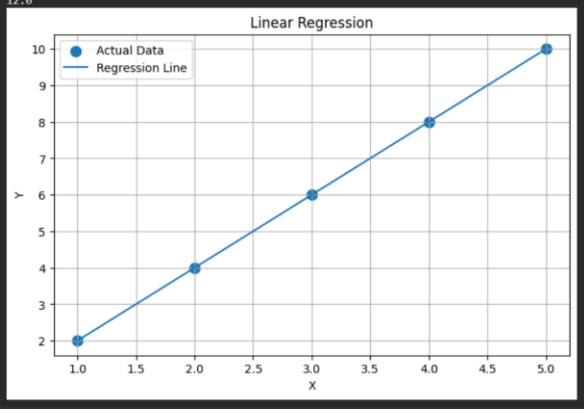
```python
plt.scatter(
    X,
    y,
    s=80,
    label="Actual Data"
)

plt.plot(
    X,
    y_pred,
    label="Regression Line"
)

plt.xlabel("X")
plt.ylabel("Y")
plt.title("Linear Regression")
plt.legend()
plt.grid(True)

plt.show()
```

```
Slope: 2.0
Intercept: 0.0
Prediction for X = 6:
12.0
```



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